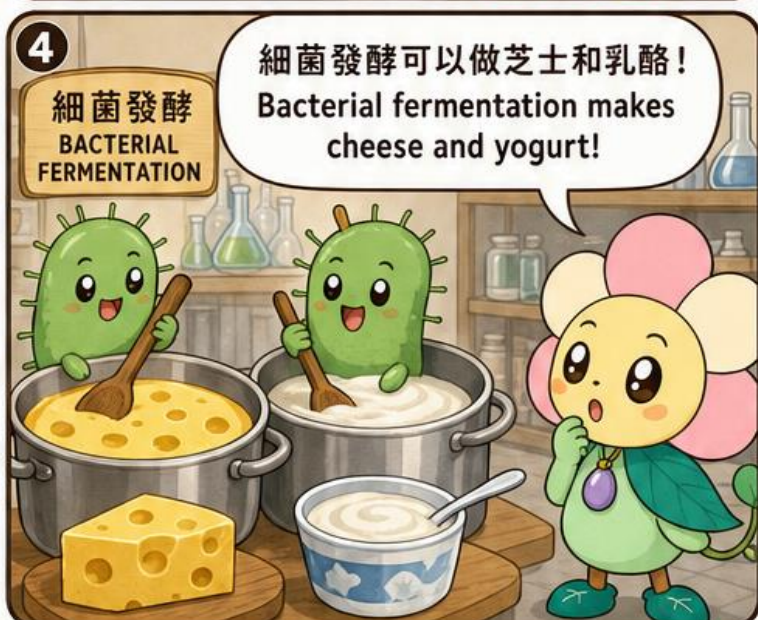
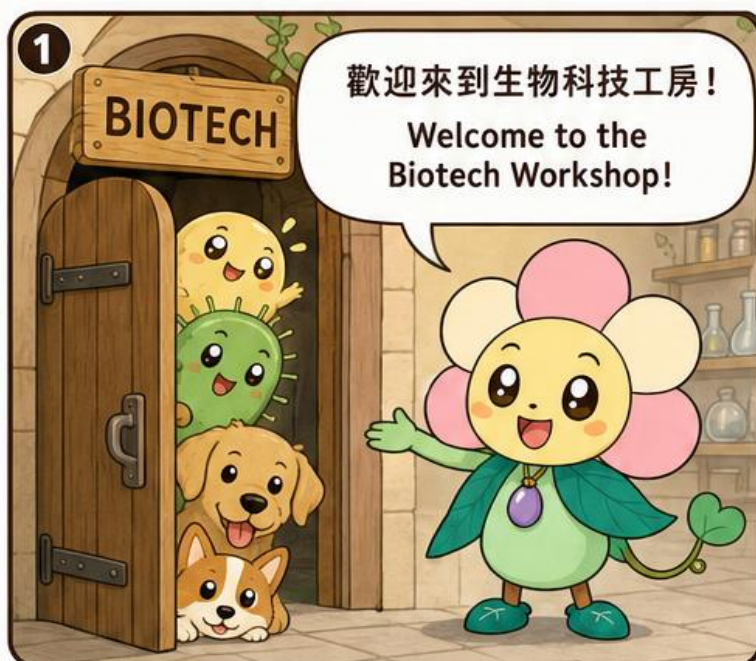
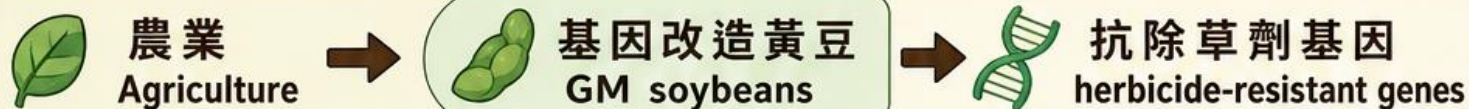






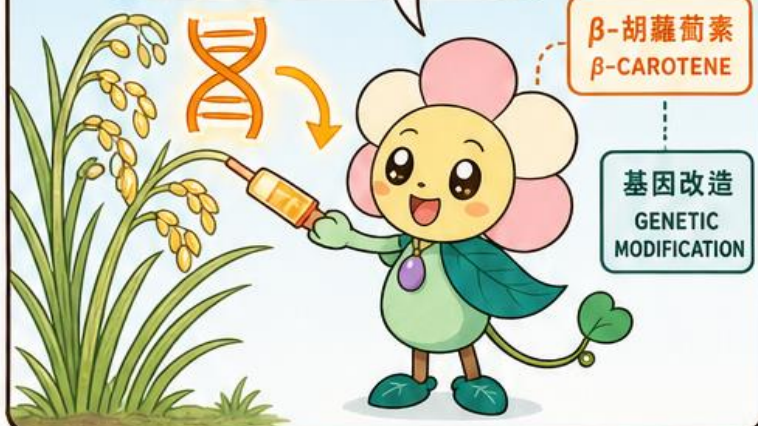
1

普通稻米缺乏維他命A。
黃金大米可以幫忙！
Ordinary rice lacks vitamin A.
Golden Rice can help!



2

把能製造β-胡蘿蔔素的
基因插入水稻。
Insert genes that produce
β-carotene into rice.



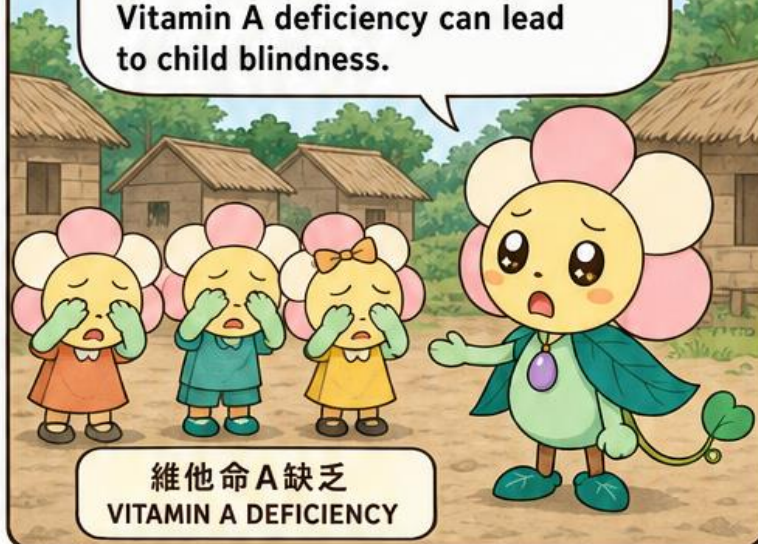
3

黃金大米不是為了好看，
而是含有β-胡蘿蔔素！
Golden Rice is not just for
colour. It contains β-carotene!



4

維他命A缺乏可導致兒童失明。
Vitamin A deficiency can lead
to child blindness.



5

黃金大米幫助發展中國家
對抗維他命A缺乏。
Golden Rice helps developing
countries fight vitamin A deficiency.



6

農業生物科技也能
保護眼睛健康！
Agricultural biotechnology
can also protect eye health!



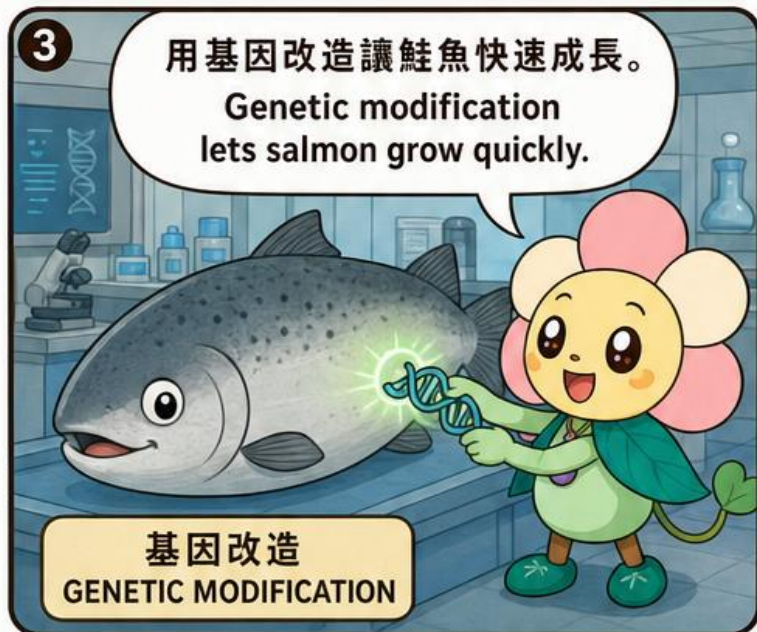
黃金大米
Golden Rice



β-胡蘿蔔素
β-carotene

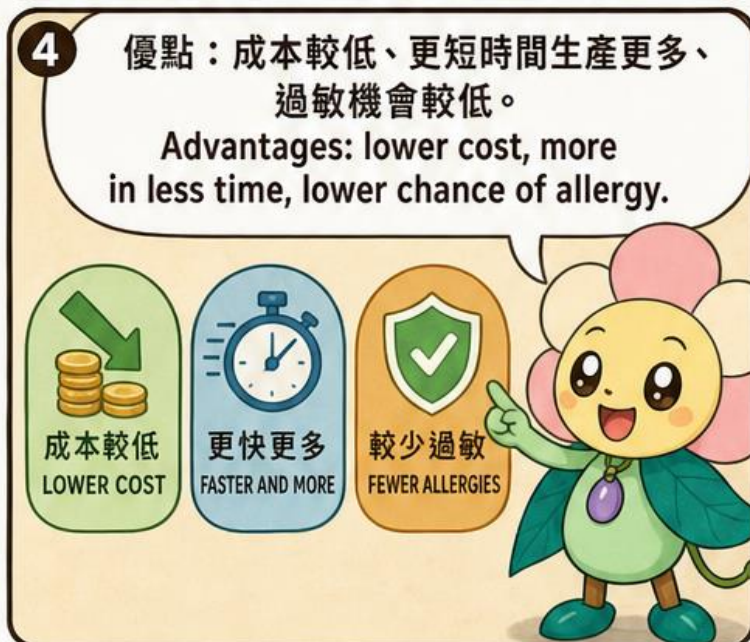
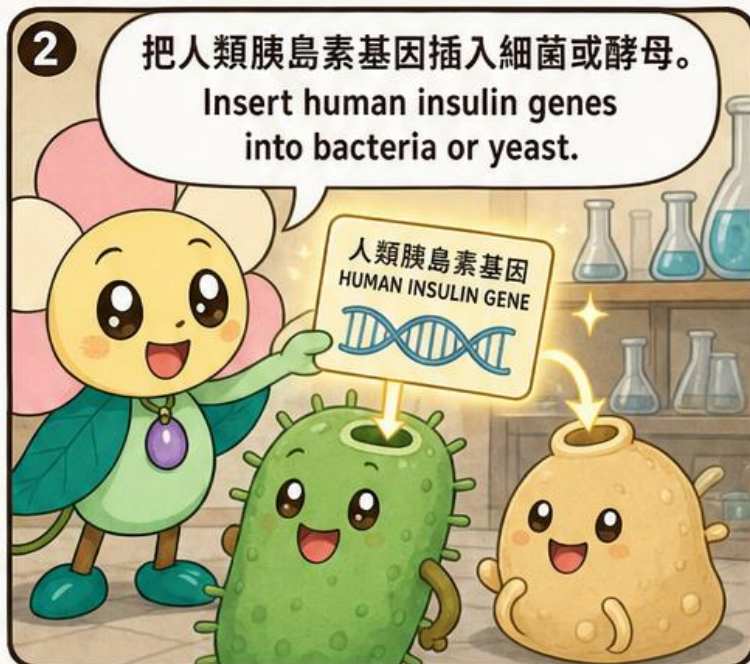
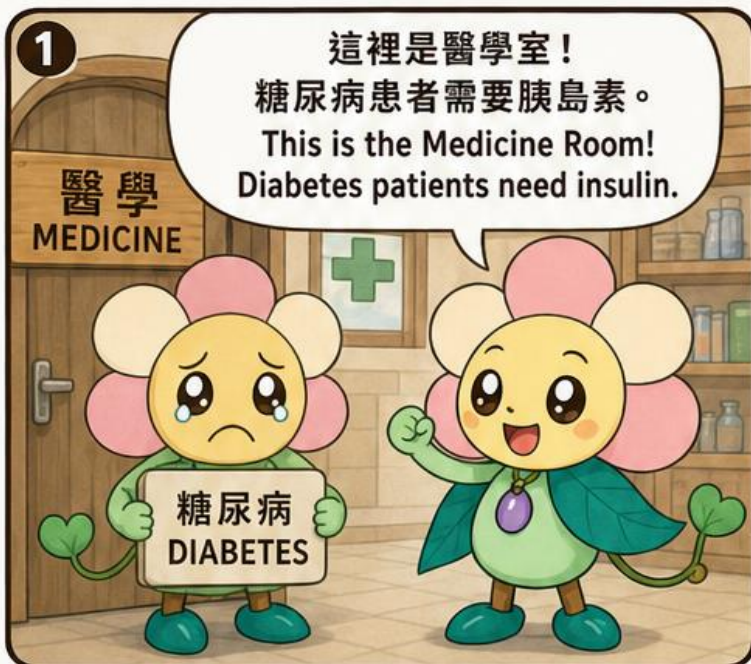


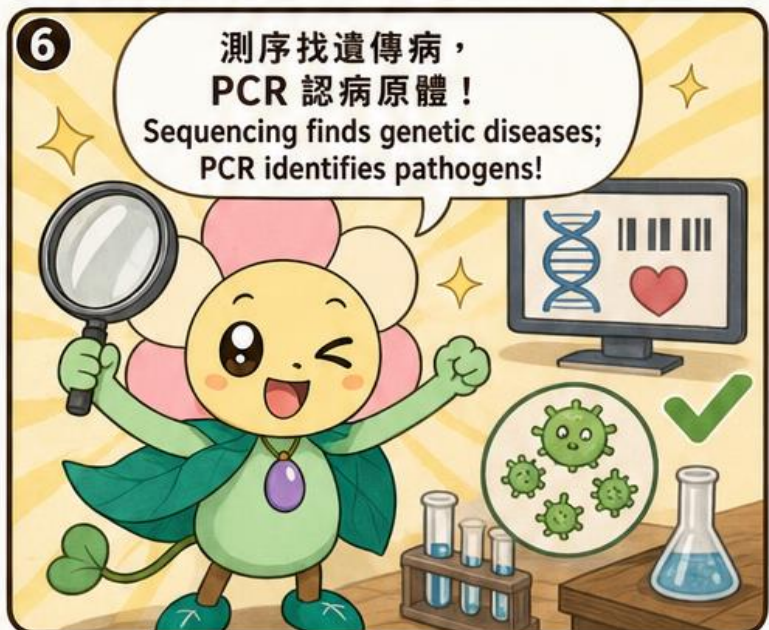
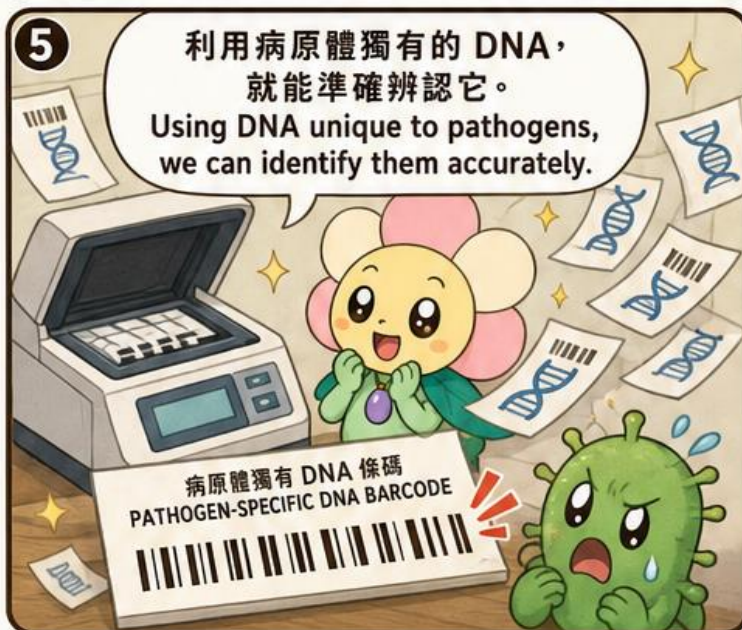
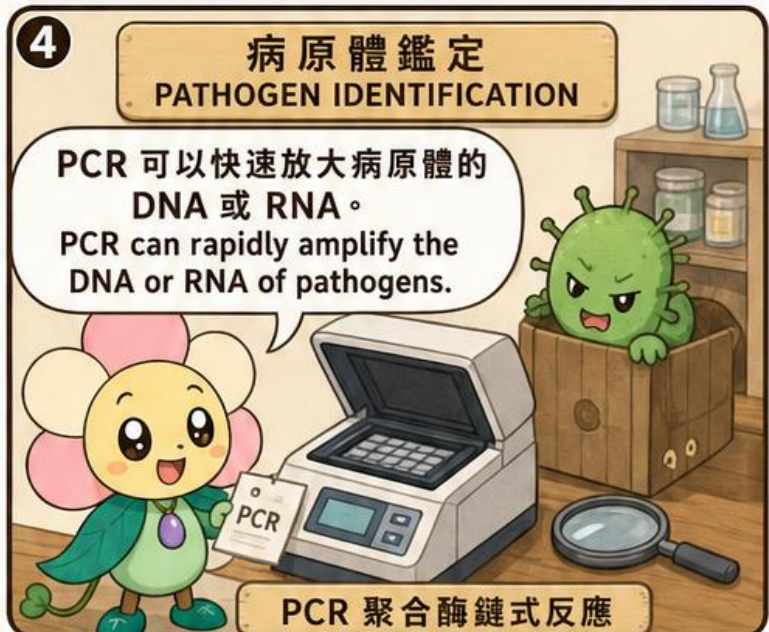
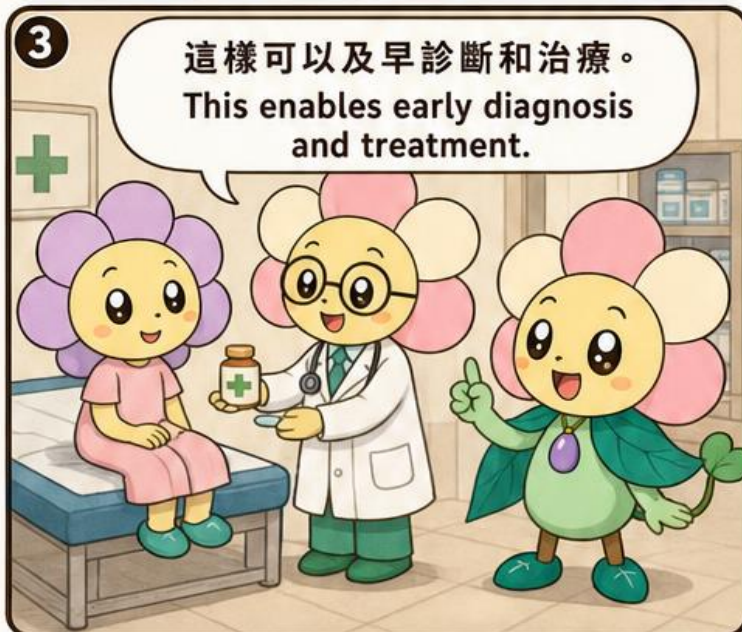
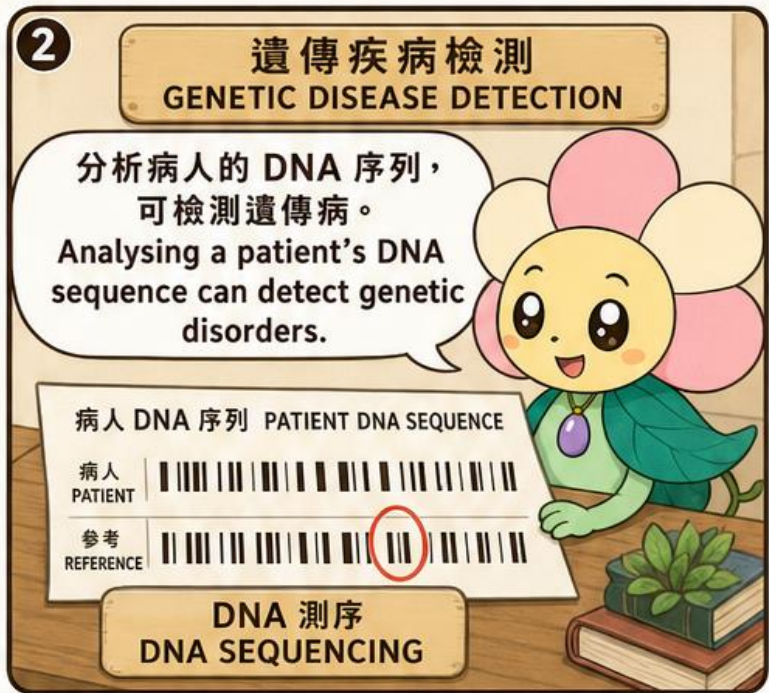
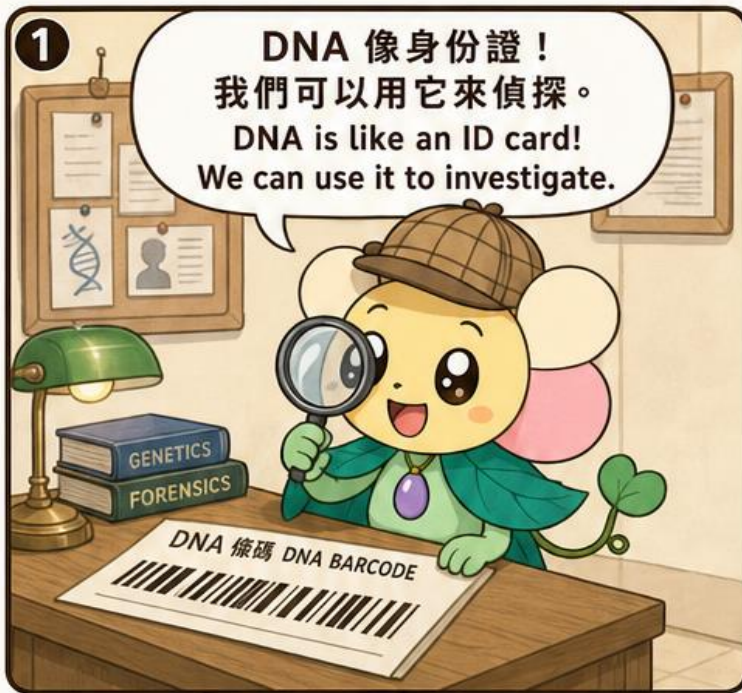
對抗維他命A缺乏
fights vitamin A deficiency



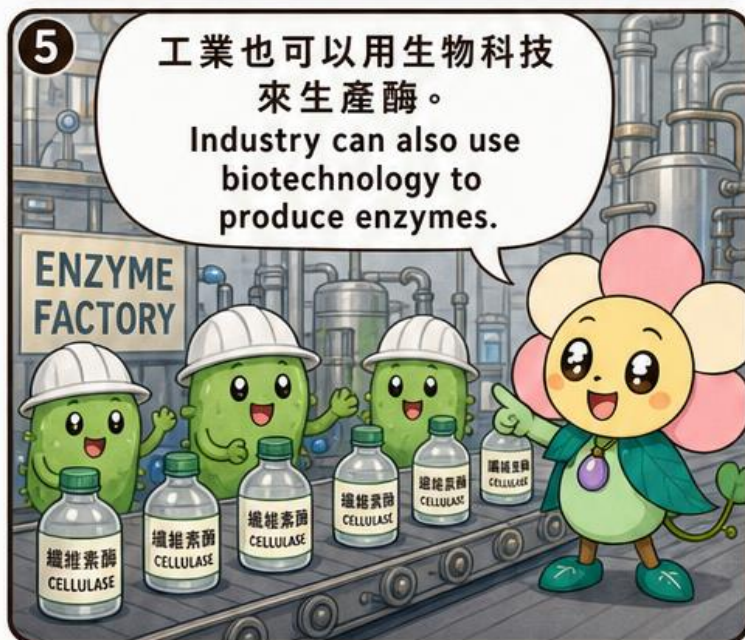
漁業 → 基因改造鮭魚 → 提高產量 /
Fisheries → GM salmon → increase yield







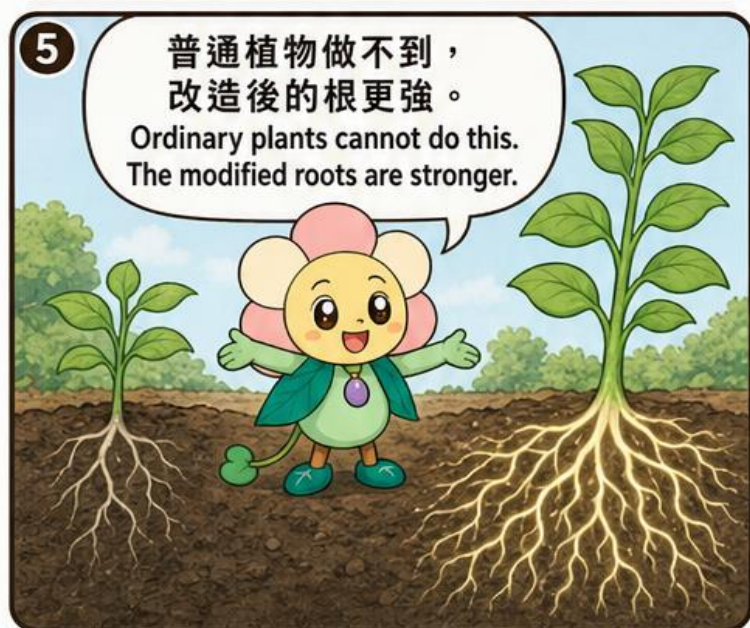
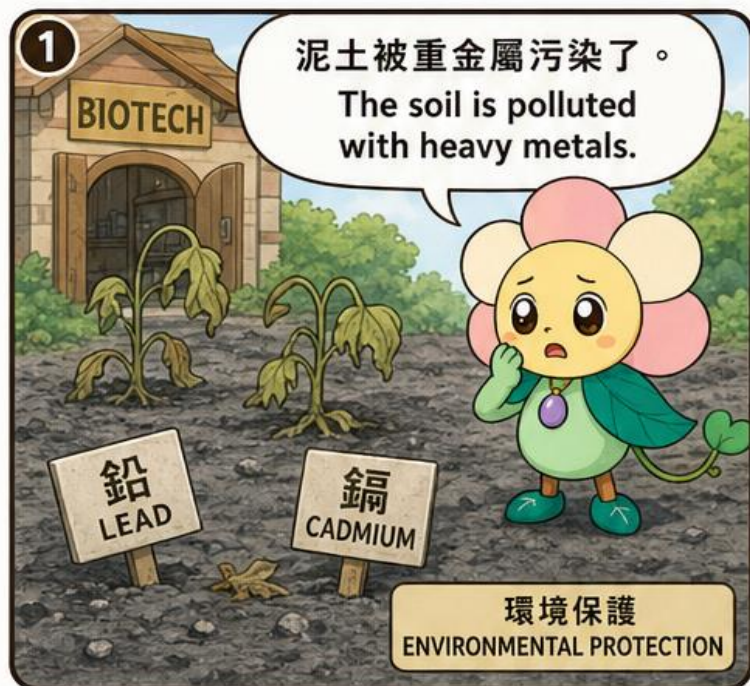
DNA 測序 + PCR → 早診斷、準確鑑定 /
DNA sequencing + PCR → early diagnosis and accurate identification



工業 → 基因改造細菌
Industry → GM bacteria

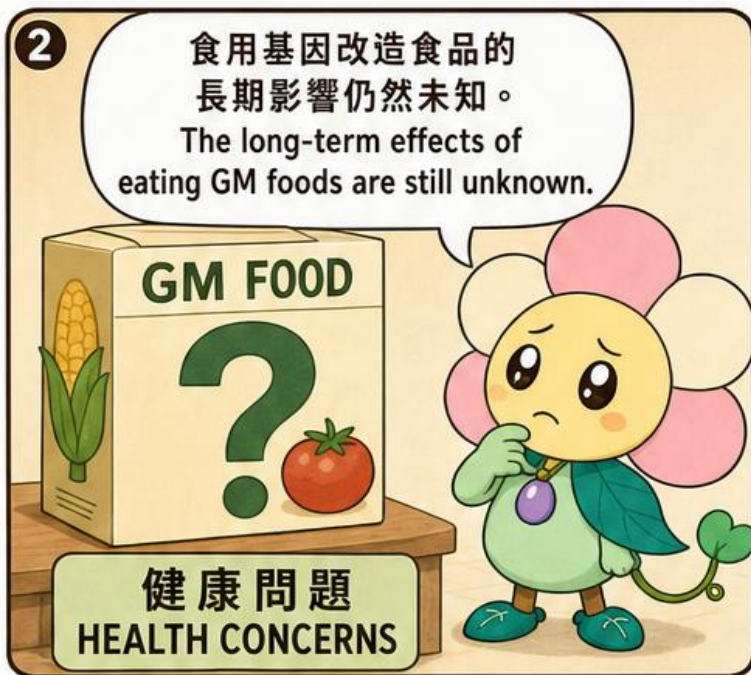
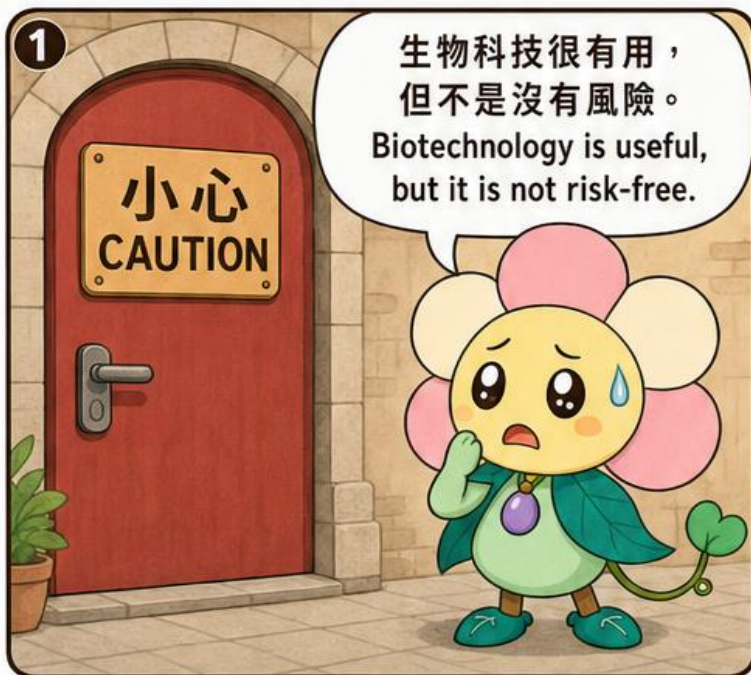
→ 纖維素酶 → 石洗
→ cellulase → stonewash





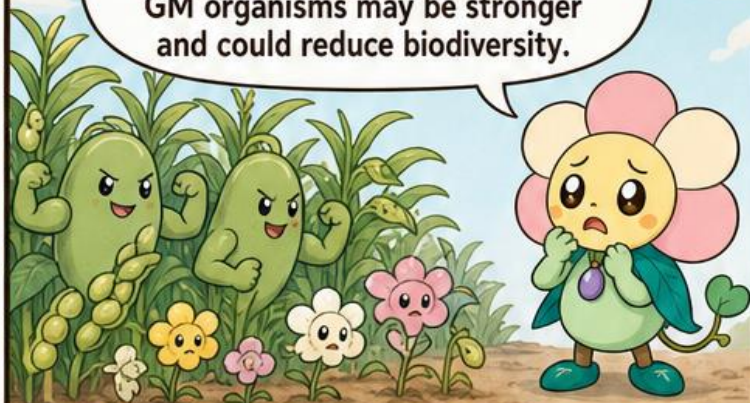
環境 → 基因改造植物 → 吸收鉛和鎘
Environment → GM plants → absorb lead and cadmium





1

基因改造生物可能更強，
會減少生物多樣性。
GM organisms may be stronger
and could reduce biodiversity.



生物多樣性
BIODIVERSITY

生態平衡
ECOSYSTEM BALANCE

2

基因污染：改造基因可能擴散到野生生物。
Gene contamination: GM genes
may spread to wild organisms.



基因污染
GENE CONTAMINATION

3

害蟲或植物可能對殺蟲劑
和除草劑產生抗性。
Pests or plants may develop
resistance to insecticides
and herbicides.



抗性問題
RESISTANCE

4

生物科技用生物幫我們種田、養魚、
治病、做布和清泥土。
Biotechnology uses living things to
farm, fish, heal, make fabric and
clean soil.



大豆
SOYBEAN

黃金稻米
GOLDEN RICE

鮭魚
SALMON

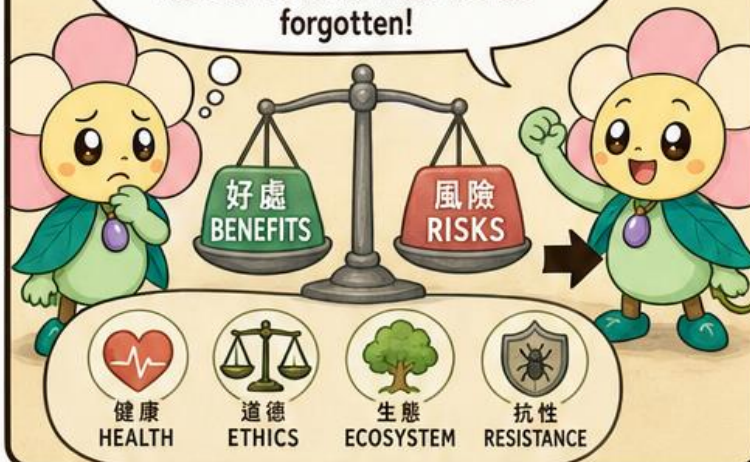
胰島素
INSULIN

布料 (牛仔褲)
JEANS

清泥土植物
CLEANING PLANT

5

健康、道德、生態和抗性風險，
都不能忘記！
Health, ethics, ecosystem and
resistance risks must not be
forgotten!



健康
HEALTH

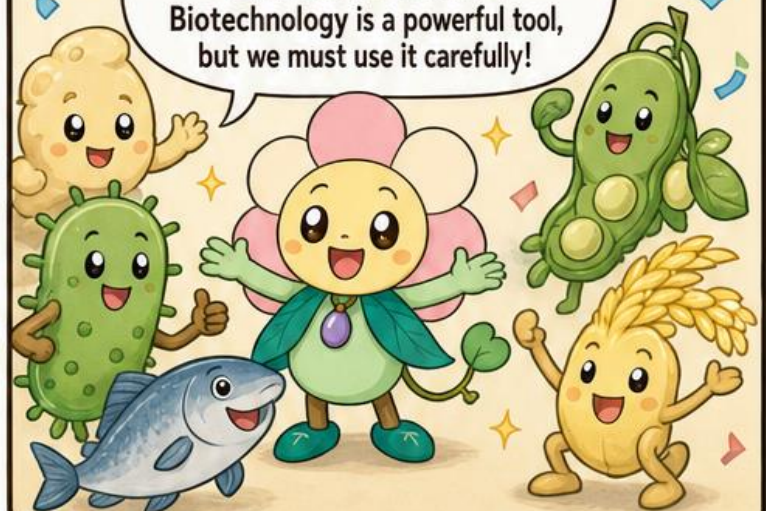
道德
ETHICS

生態
ECOSYSTEM

抗性
RESISTANCE

6

生物科技是強大工具，
但我們要小心使用！
Biotechnology is a powerful tool,
but we must use it carefully!



有用 + 小心風險 = 負責任的生物科技

Useful + careful of risks = responsible biotechnology